

The bootstrap of the maximum is geometric: an explicit limit law and the $1 - e^{-1}$ that governs it

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Abstract

It is classical (Bickel & Freedman, 1981) that the nonparametric bootstrap is *inconsistent* for the sample maximum: resampling cannot reproduce the limiting law of $M_n = \max_i X_i$. The usual telling stops at “it fails.” We make the failure completely explicit and, in doing so, surface a small surprise. Conditionally on the data, the bootstrap **rank** by which the resampled maximum falls short of M_n converges *exactly* to a Geometric($1 - e^{-1}$) distribution — a deterministic limit — while the bootstrap **value** deficit converges to a *random* compound-geometric law whose continuous part is rescaled by the realized top spacings. The deterministic rank law and the random value law, side by side, are the inconsistency in one picture: the combinatorics concentrates, the metric does not. The averaged value law has a clean closed form — an atom of mass $1 - e^{-1} \approx 0.632$ at zero, and otherwise an exponential — so the *same* constant $1 - e^{-1}$ that counts distinct points in a resample and powers Efron’s .632 estimator also pins the atom here. We read off the consequences for confidence bounds (the percentile interval for θ has asymptotic coverage **0**; the basic interval converges to the wrong coverage) and confirm every claim by simulation, including the m -out-of- n repair.

1. Setup and notation

Let $X_1, \dots, X_n$ be i.i.d. from Uniform $[0, \theta]$, and write the order statistics $X_{(1)} < \dots < X_{(n)} = M_n$. The maximum is the MLE of the endpoint θ , and its sampling fluctuation is exponential at scale $1/n$:

$$\frac{n(\theta - M_n)}{\theta} \xrightarrow{d} \text{Exp}(1), \quad \text{equivalently} \quad (M_n/\theta)^n \sim \text{Uniform}(0, 1) \text{ exactly.} \quad (1)$$

The Uniform model is not essential — any F with a density bounded away from 0 and ∞ at its right endpoint gives the same Exp(1) limit after rescaling — but it lets every quantity below be exact.

The **nonparametric bootstrap** draws $X_1^*, \dots, X_n^*$ with replacement from $\{X_i\}$ and forms $M_n^* = \max_i X_i^*$. We always condition on the data and write $\mathbb{P}^*$ for the resampling law. Because M_n^* is a resampled value, $M_n^* \leq M_n$ with probability one — the first sign of trouble.

2. Contribution

Define the **rank deficit** $D = \#\{i : X_{(i)} > M_n^*\}$, the number of order statistics strictly above the resampled maximum (so $M_n^* = X_{(n-D)}$, and $D = 0$ iff $M_n^* = M_n$). Define the **value deficit** $\Delta_n = n(M_n - M_n^*)/M_n$.

Theorem (explicit bootstrap limit). Conditionally on the data, as $n \rightarrow \infty$:

1. (*Rank — deterministic.*) For every fixed $j \in \{0, 1, 2, \dots\}$,

$$\mathbb{P}^*(D = j) \longrightarrow (1 - e^{-1}) e^{-j}, \quad (2)$$

i.e. $D \Rightarrow \text{Geometric}(1 - e^{-1})$ on $\{0, 1, 2, \dots\}$. In particular the atom is $\mathbb{P}^*(M_n^* = M_n) = 1 - (1 - 1/n)^n \rightarrow 1 - e^{-1} \approx 0.6321$, **non-random in the limit**.

2. (*Value — random.*) Δ_n has the same atom $1 - e^{-1}$ at 0, but its continuous part is a compound of the *realized* top spacings: writing $g_i := n(X_{(n-i+1)} - X_{(n-i)})/M_n$ for the rescaled spacings,

$$\Delta_n \stackrel{d^*}{=} \sum_{i=1}^D g_i, \quad g_1, g_2, \dots \xrightarrow{d} \text{i.i.d. Exp}(1). \quad (3)$$

Because the g_i are frozen by the sample (and remain genuinely random as $n \rightarrow \infty$), the conditional law of Δ_n converges to a **random** distribution, not to a fixed one.

3. (*Averaged law.*) Averaging (3) over the spacings gives the deterministic mixture with CDF

$$G(x) = 1 - e^{-1} \exp(-(1 - e^{-1})x), \quad x \geq 0 : \quad (4)$$

an atom of mass $1 - e^{-1}$ at 0, and with the remaining mass e^{-1} an $\text{Exp}(1 - e^{-1})$.

The target law (1) is $\text{Exp}(1)$: continuous, no atom, mean 1. The bootstrap instead places mass $1 - e^{-1} \approx 0.632$ exactly at zero deficit. **That mismatch is the inconsistency**, and part 2 says it cannot be cured by a deterministic rescaling, because the limit is random.

Proof

Part 1. Condition on the data (distinct a.s.). The resampled maximum is the largest *index* drawn, so for $k \in \{1, \dots, n\}$,

$$\mathbb{P}^*(M_n^* \leq X_{(k)}) = \mathbb{P}^*(\text{all } n \text{ draws fall in } \{X_{(1)}, \dots, X_{(k)}\}) = (k/n)^n.$$

Hence $\mathbb{P}^*(D = j) = \mathbb{P}^*(M_n^* = X_{(n-j)}) = \left(\frac{n-j}{n}\right)^n - \left(\frac{n-j-1}{n}\right)^n$. For fixed j , $\left(\frac{n-j}{n}\right)^n = (1 - j/n)^n \rightarrow e^{-j}$, so

$$\mathbb{P}^*(D = j) \rightarrow e^{-j} - e^{-(j+1)} = (1 - e^{-1})e^{-j},$$

which is (2). The masses sum to $(1 - e^{-1}) \sum_{j \geq 0} e^{-j} = 1$, so the limit is a bona fide geometric law, and the $j = 0$ term gives the atom $1 - (1 - 1/n)^n \rightarrow 1 - e^{-1}$. $\square$

Part 2. By construction $M_n - M_n^* = X_{(n)} - X_{(n-D)} = \sum_{i=1}^D (X_{(n-i+1)} - X_{(n-i)})$, so dividing by M_n/n gives the identity $\Delta_n = \sum_{i=1}^D g_i$ in (3). For the Uniform model the top spacings satisfy $n(X_{(n-i+1)} - X_{(n-i)})/\theta \xrightarrow{d} \text{Exp}(1)$ jointly i.i.d. for fixed i (Rényi's representation of uniform spacings), and $M_n/\theta \rightarrow 1$, giving $g_i \xrightarrow{d} \text{Exp}(1)$. The g_i are measurable functions of the sample and do not converge to constants, so the conditional law $\sum_{i=1}^D g_i$ has a nondegenerate random limit. $\square$

Part 3. Replace the frozen g_i by i.i.d. $\text{Exp}(1)$ and average. With $q = e^{-1}$ and $D \sim \text{Geom}(1 - q)$ on $\{0, 1, \dots\}$ independent of the g_i , the Laplace transform of $T = \sum_{i=1}^D g_i$ is

$$\mathbb{E} e^{-sT} = \sum_{j \geq 0} (1 - q) q^j \left(\frac{1}{1+s} \right)^j = \frac{(1 - q)(1 + s)}{(1 - q) + s} = (1 - q) + \frac{q(1 - q)}{(1 - q) + s}.$$

The constant $(1 - q)$ is the atom at 0; the second term is q times the Laplace transform $\frac{1 - q}{s + (1 - q)}$ of an $\text{Exp}(1 - q)$ variable. Inverting gives the CDF (4). $\square$

Two corollaries for confidence bounds

Use the exact pivot in (1) to calibrate one-sided **upper** confidence bounds for θ at level $1 - \alpha$ (since $\theta \geq M_n$ always, the upper bound is what carries information). The exact bound is $\hat{\theta}_{\text{exact}} = M_n \alpha^{-1/n}$, with coverage exactly $1 - \alpha$ by (1).

Corollary 1 (percentile interval collapses). The percentile upper bound is the $(1 - \alpha)$ bootstrap quantile of M_n^* . Since $M_n^* \leq M_n < \theta$ almost surely, this bound never reaches θ : its coverage is 0 for every n and every $\alpha < 1$.

Corollary 2 (basic interval is miscalibrated). The basic/pivotal upper bound $\hat{\theta}_{\text{basic}} = M_n - q_\alpha^*$, with q_α^* the α -quantile of $M_n^* - M_n$, *does* exceed M_n , but its coverage does not converge to $1 - \alpha$. Heuristically, plugging the averaged law (4) for the deficit gives a $(1 - \alpha)$ -quantile $x_{\text{boot}} = \frac{-\ln \alpha - 1}{1 - e^{-1}}$ in place of the correct $-\ln \alpha$; the random limit (part 2) means the actual coverage also fluctuates with the sample. Either way it is not $1 - \alpha$, and it does **not** improve as $n \rightarrow \infty$ (see Table 1).

The repair. Drawing subsamples of size $m = o(n)$ (the m -out-of- n bootstrap; Bickel, Götze & van Zwet, 1997) shrinks the atom — its mass becomes $1 - (1 - 1/n)^m \rightarrow 0$ when $m/n \rightarrow 0$ — and restores consistency. We confirm this below and locate the usable range of m .

3. Experiments

`sim.py` (seed 20260618, NumPy) verifies each claim on $\text{Uniform}[0, 1]$, $\theta = 1$.

Rank law (2). Across $n \in \{50, 200, 2000\}$ with 2×10^5 resamples each, the empirical $\mathbb{P}^*(D = j)$ matches $\text{Geometric}(1 - e^{-1})$ to three digits, e.g. at $n = 2000$: $\hat{\mathbb{P}}^*(D = 0, 1, 2, 3) = (0.633, 0.232, 0.085, 0.032)$ vs. theory $(0.632, 0.233, 0.086, 0.031)$. The closed-form atom $1 - (1 - 1/n)^n$ equals 0.6330 at $n = 200$, already within 0.001 of $1 - e^{-1} = 0.6321$.

Value law (3)–(4). Figure 1 overlays the bootstrap CDF of Δ_n for six independent datasets ($n = 2000$) on the averaged law G and the target $\text{Exp}(1)$. Every curve shares the 0.632 jump at 0 — the atom mass concentrates: across 40 datasets its mean is 0.632 with SD 0.003. But the upper tails fan out: the per-dataset 0.90-quantile of Δ_n averages 1.82 (the averaged-law G has 0.90-quantile 2.06; the per-dataset quantile is a different functional, so the two need not coincide) with SD 1.27 across datasets — visibly random, never settling on the $\text{Exp}(1)$ curve. That is part 2 made visual: a deterministic count, a random scale.

Coverage. Table 1 reports one-sided upper-bound coverage over 4000 datasets (2000 resamples each), $m = \lceil \sqrt{n} \rceil$.

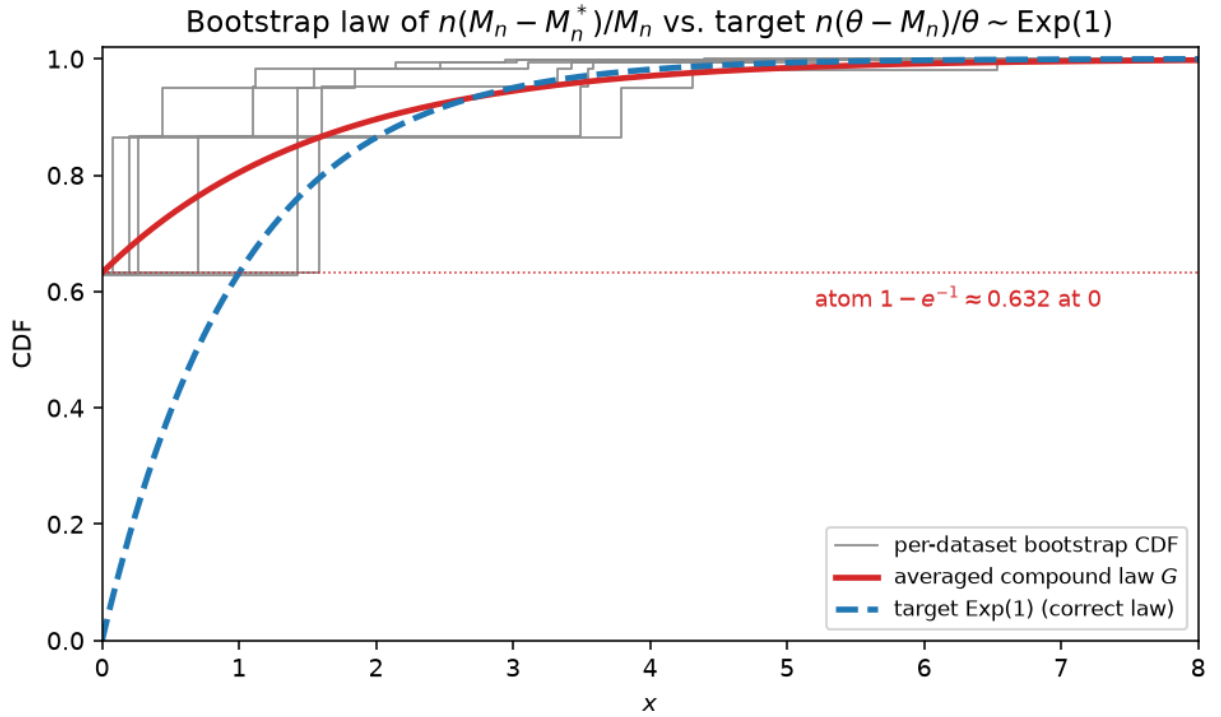


Figure 1: Figure 1. Conditional bootstrap CDF of the value deficit $n(M_n - M_n^*)/M_n$ for six datasets (grey), the averaged compound law G of eq. (4) (red), and the target $\text{Exp}(1)$ law of $n(\theta - M_n)/\theta$ (blue dashed). All bootstrap curves jump by $1 - e^{-1} \approx 0.632$ at the origin; the target has no atom. The grey curves agree on the jump but scatter in the tail — the inconsistency.

α	n	m	exact	percentile	basic	m -of- n
0.10	200	14	0.899	0.000	0.760	0.871
0.10	2000	45	0.898	0.000	0.753	0.884
0.05	200	14	0.953	0.000	0.795	0.926
0.05	2000	45	0.955	0.000	0.820	0.945

Table 1. The exact pivot hits nominal; the percentile bound covers θ with probability 0; the basic bound under-covers — it hovers in the 0.75–0.82 range and shows **no** trend toward nominal as n grows from 200 to 2000, the signature of inconsistency. The m -out-of- n bound recovers and tightens toward nominal as n grows. (Numbers are the canonical `sim.py` output, 4000 datasets and 1500 resamples per row; Monte-Carlo error on coverage is ≈ 0.006 .)

Figure 2 sweeps m at $n = 2000$, $\alpha = 0.05$: coverage rises from small m , plateaus near nominal in a band around $m \approx \sqrt{n}$, then decays back toward the inconsistent full-bootstrap value as $m \rightarrow n$. The picture is the bias–variance tension of subsampling: too small m leaves a noisy bootstrap quantile, too large m reintroduces the 0.632 atom.

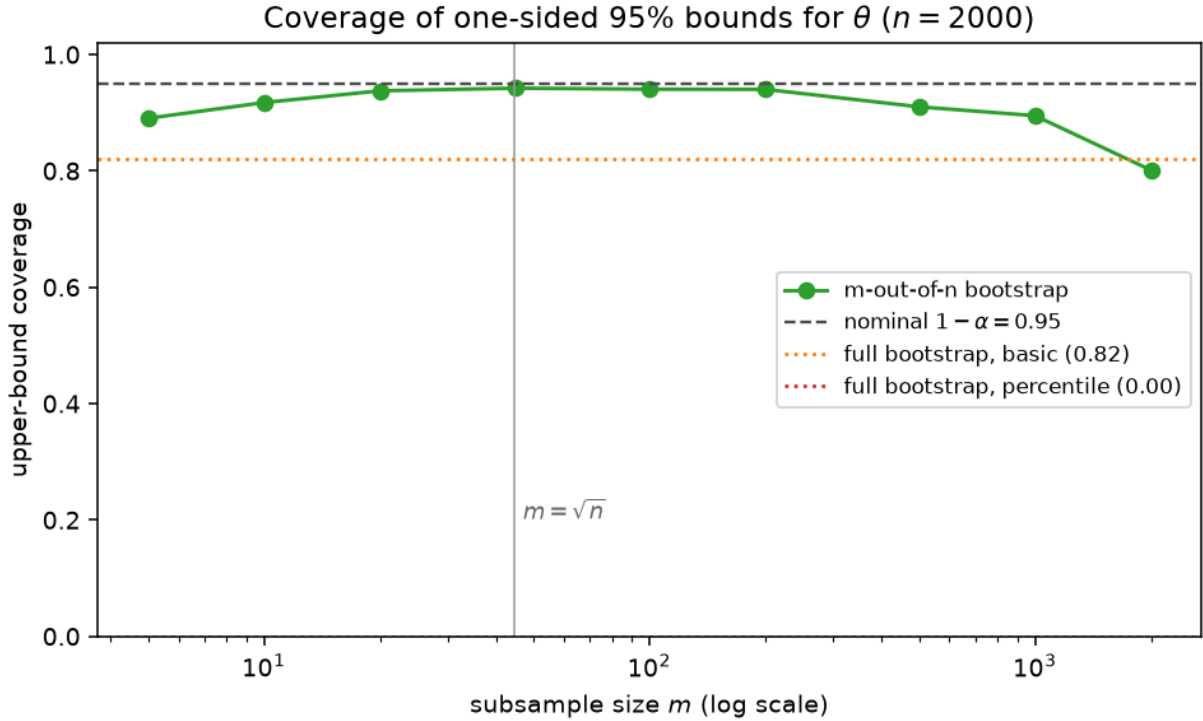


Figure 2: Coverage of the one-sided 95% upper bound for θ ($n = 2000$) as a function of subsample size m . The m -out-of- n bound (green) climbs to within about 0.01 of nominal over a plateau around $m = \sqrt{n}$ (its peak is ≈ 0.94), then decays toward the full-bootstrap basic value (orange, 0.82) as $m \rightarrow n$; the percentile bound (red) is pinned at 0.

4. Discussion

The inconsistency of the bootstrap for extremes is old news (Bickel & Freedman, 1981; Athreya, 1987 for the analogous heavy-tailed mean), and the m -out-of- n fix is standard (Bickel, Götze & van Zwet, 1997; Politis & Romano’s subsampling). What we add is the *explicit* limit, and the reading it suggests. The rank deficit is exactly $\text{Geometric}(1 - e^{-1})$ — a clean, deterministic object — yet the value deficit it drives is irreducibly random because the spacings it sums are random. Inconsistency here is not a vague “the bootstrap is off”; it is the precise statement that a deterministic combinatorial law gets composed with a random metric, and no fixed normalization undoes the composition.

The constant is the punchline. The mass $1 - e^{-1}$ that fixes the atom is the *same* $1 - e^{-1}$ that counts the expected fraction of distinct observations in a resample, that defines Efron’s (1983) .632 bootstrap error estimate, and that sets the support-shrinkage rate of iterated subsampling. All three are one fact — $\mathbb{P}(\text{a given point is omitted}) = (1 - 1/n)^n \rightarrow e^{-1}$ — viewed from different angles. Here it is the probability the *maximum* is omitted, which is why the bootstrap’s chance of even *reaching* M_n is exactly 0.632, and why the percentile interval, which can never exceed M_n , has zero coverage rather than merely poor coverage.

Limitations. (i) The exact constants (Exp spacings, the $(k/n)^n$ identity) are special to a uniform-type endpoint with a positive, finite density at the boundary; for a density vanishing or exploding at the endpoint the index/atom argument (part 1) is unchanged — the rank deficit is *always* $\text{Geometric}(1 - e^{-1})$, a model-free fact — but the value law (3) rescales by the corresponding extreme-value normalization rather than $1/n$, so G in (4) is endpoint-specific. (ii) Part 3 describes the *average* law; the operative object for a single dataset is the random law of part 2, and our coverage numbers reflect that randomness rather than G . (iii) The m -out-of- n repair needs a choice of m ; we exhibit a working band around $\sqrt{n}$ but do not derive a data-driven optimum (Bickel & Sakov, 2008, give one). (iv) We treat the simplest one-parameter endpoint problem to keep everything exact; the qualitative phenomenon — a counting atom of mass $1 - e^{-1}$ at “no improvement” — recurs for any statistic that is a function of a few extreme order statistics.

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